

Arun Madhusudhanan

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EDUCATION

Master of Science in Robotics, Computer Science Concentration

Sep 2022 - Dec 2024

Northeastern University, Boston, MA

GPA: 4.0/4.0

Courses: Computer Vision, Robotic Sensing and Navigation, Reinforcement Learning, Artificial Intelligence

Bachelor of Technology in Mechanical Engineering, NIT Calicut, India

Jul 2014 - May 2018

TECHNICAL SKILLS

- **Programming Languages:** C++, Python
- **Libraries:** PyTorch, OpenCV, PCL, Open3D, TensorRT, ONNX, Scikit-learn, NumPy, OpenVINO
- **Software Tools:** MATLAB, ROS, Docker, Git, Ubuntu, AWS ECS, FastAPI, OriginPro, SolidWorks

PROJECTS

Open Vocabulary 3D Scene Understanding

Nov 2024 - Dec 2024

- Designed a **multimodal 3D scene understanding** pipeline capable of processing audio, text, and image inputs, enabling robots to interpret complex natural language commands and perform actions.
- Leveraged vision foundation models (**RAM, DINO, SAM, CLIP**) for object detection, segmentation, and feature extraction from 2D images, and merged the results onto point clouds for **3D semantic segmentation**.
- Incorporated **scene graphs** with object-room hierarchical relationships and used the **Qwen2.5-72B-Instruct** model for text query parsing, improving **natural language object retrieval accuracy by 95%** on the Replica dataset.
- Integrated **GraspNet** for **Open-Vocabulary Multi-Modal Grasping**, enabling robots to generalize grasping with natural language commands, with simulations done in **PyBullet**. [\[Code\]](#)

Time-to-Collision (TTC) Estimation using LiDAR and Camera [\[Code\]](#)

Jul 2024 - Aug 2024

- Developed a **LiDAR and Camera** based Time-to-Collision estimation pipeline by clustering **point clouds** using **YOLOv8**-detected 2D bounding boxes and tracking objects via **keypoint matching**.
- Benchmarked **OpenCV-DNN, ONNX-Runtime and OpenVINO** for YOLOv8 integration in C++ & CPU deployment.
- **Achieved a 3× speedup** in YOLOv8 inference by integrating OpenVINO and implementing **static INT8 quantization**.

Other Academic Projects

Dec 2022 - Oct 2024

- Developed a **3D object classification pipeline** for incomplete point clouds using **GRNet** for completion and **PointNet** for classification, achieving **93.8%** accuracy, outperforming PointNet++'s 70%. [\[Code\]](#) [\[Report\]](#)
- Optimized **YOLOv8 with NVIDIA TensorRT**, achieving real-time inference speeds of up to 60 FPS. [\[Code\]](#) [\[Report\]](#)
- Implemented Deep Learning Models from scratch: **Tiny-NeRF** for novel view synthesis, **Stable Diffusion**, **PointNet** for point cloud processing, **Visual Transformer**, **GAN & VAE**. [\[Code\]](#)
- Developed a scalable **FastAPI** Image Classification Application using **AWS ECS** and **Docker** containerization. [\[Code\]](#)
- Fine-tuned **ViT+GPT-2** for **image captioning** and compared performance with a **ResNet+LSTM** model. [\[Code\]](#)
- Built a robust **Vehicle Tracking system** by fusing **LiDAR and Radar** data with **Unscented Kalman Filter**. [\[Code\]](#)
- Built a **Structure from Motion** pipeline in C++ with **Bundle Adjustment** for **3D reconstruction** from images. [\[Code\]](#)
- Developed **monocular Visual Odometry** system in C++ featuring **pose graph optimization** using **GTSAM** and GPS integration for scale estimation. [\[Code\]](#)

WORK EXPERIENCE

Graduate Teaching Assistant, Computer Vision

Jan 2024 - Apr 2024

Northeastern University, Boston, MA

- Reviewed code, debugged issues, and graded projects in C++, Python, OpenCV, and PyTorch for 120+ students.
- Mentored students on topics such as image processing, object recognition, camera calibration, key points, stereo vision.

Machine Learning Research Intern

Jul 2023 - Dec 2023

Festo Corporation, Marlborough, MA

- Developed a robust machine learning pipeline to predict the output parameters of a high-precision liquid dosing unit achieving an error rate of less than 2.5%; the device with incorporated ML model is currently under patent application.
- Managed all aspects of the ML pipeline, from dataset collection and feature extraction to data labeling, model development, and live inference.
- Optimized software modules for sensor activation, hardware drivers, and data storage by refactoring code, fixing bugs, and implementing multi-threading, resulting in a significant reduction in system latency.

Wells Engineer

Jul 2018 - Jun 2022

ExxonMobil, Bengaluru, India

- Led beta testing of company-wide tubular design software, ensuring robustness and reliability before release to users.
- Stewarded and improved the tubular design workflow for business divisions across the world in accordance with industry standard API 5C5, resulting in \$230k immediate savings and long-term synergistic benefits.